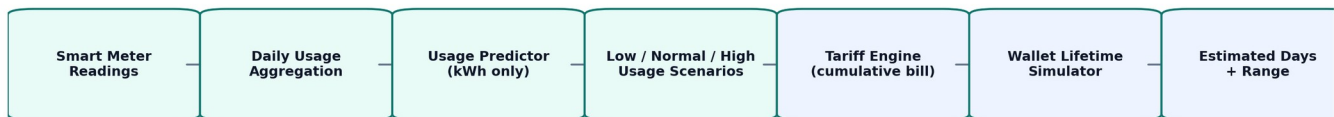
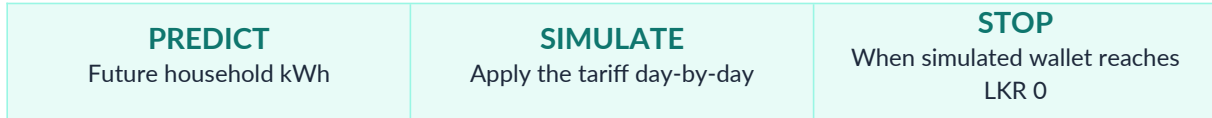


Estimated Days Remaining

Technical Design Guide for Wallet Lifetime Estimation

How to estimate how many days a prepaid balance can power a household under a nonlinear electricity tariff.



Prediction estimates future kWh. The tariff engine alone converts kWh into LKR.

Core principle

Predict future electricity usage in kWh first. Then use the same deterministic tariff engine used for real billing to convert that usage into LKR. Never predict wallet lifetime from one fixed LKR/kWh rate.

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1. Why the Simple Formula Is Not Enough

Do not use

Days = Wallet Balance / (Average Daily kWh x Current Tariff Rate)

That formula assumes every future unit of electricity has the same price. In the domestic tariff, the effective cost can change when the billing-period consumption crosses a threshold. Some tariff structures also recalculate earlier units under a different group once a threshold is crossed.

- The next 1 kWh can cost differently depending on current billing-cycle usage.
- Crossing a tariff threshold can create a larger one-time increase in cumulative bill liability.
- A future billing-cycle reset changes the tariff calculation back to the beginning of a new cycle.
- Fixed charges must be applied consistently and only through the official tariff calculation logic.
- A wallet may last across more than one billing period, so estimation must simulate future dates, not only future energy.

The correct mental model

The estimator answers one question: "If this household continues using electricity in a realistic way, on which future day will the prepaid wallet be unable to cover the cumulative tariff charge?"

2. Data the Estimator Needs

Input	Example	Why it matters
Wallet balance	LKR 3,970.00	Amount available to pay future electricity charges.
Current cycle usage	82.0 kWh	Determines the current tariff position. This is NOT the lifetime meter reading.
Billing-cycle dates	Start / end date	Needed to know when usage resets to zero for tariff purposes.
Daily usage history	7-30+ days	Used to estimate future household consumption.
Active tariff version	Version ID + dates	Ensures the correct tariff is used for each simulated future date.
Previously charged cycle amount	LKR value	Lets the system deduct only the new incremental liability.

Critical distinction

Keep lifetime cumulative meter reading and current billing-cycle consumption as separate fields. The tariff estimator must use current billing-cycle consumption.

3. Phase 1 Usage Prediction - No ML Required

For the first implementation, use an explainable historical predictor. This is easier to test, cheap to run, and provides a strong baseline before introducing machine learning.

Step A - Recent weighted average

```
recentAverage =  
    0.50 x last7DayAverage  
+ 0.30 x last14DayAverage  
+ 0.20 x last30DayAverage
```

Step B - Day-of-week behavior

Calculate the average usage of the same weekday from recent weeks. For example, when predicting a Sunday, use the previous Sundays when available.

```
predictedDailyKWh =  
    0.60 x recentAverage  
+ 0.40 x sameWeekdayAverage
```

Fallback rules

Available history	Prediction method
30+ days	Full 7/14/30-day weighted model + same weekday
14-29 days	Use available 7/14-day averages + weekday pattern
7-13 days	Use 7-day average, optionally adjusted by weekday
1-6 days	Use average of available complete days
No reliable history	Return estimate unavailable; do not invent a value

Return a range, not false precision

Run the same wallet simulation under three usage scenarios:

Scenario	Daily usage multiplier	Meaning
Low usage	0.85 x prediction	Longer wallet lifetime
Normal usage	1.00 x prediction	Main estimate
High usage	1.15 x prediction	Shorter wallet lifetime

4. The Recommended Billing Method: Cumulative Liability

Recommended implementation

At every meter update, calculate the total tariff charge that should have accrued for the current billing period at the current total kWh. Then deduct only the difference from what has already been charged for that cycle.

```
currentBillLiability = tariffEngine.calculateBill(cycleKWh, date)

incrementalDebit = currentBillLiability - previouslyChargedForCycle

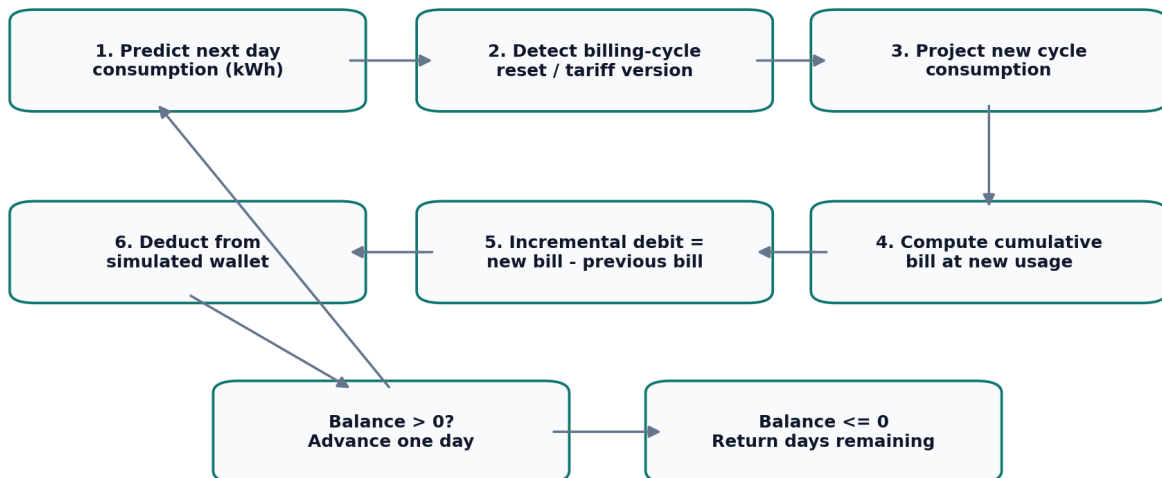
walletBalance = walletBalance - incrementalDebit
previouslyChargedForCycle = currentBillLiability
```

This is safer than manually choosing a current slab rate. It automatically captures threshold crossings, retroactive group changes, and fixed-charge changes as long as `tariffEngine.calculateBill()` is the single source of truth.

Why this matters for estimation

The future wallet simulator should use the exact same function. For each simulated day, it projects the new cycle consumption, asks the tariff engine for the new cumulative bill liability, and subtracts only the increase from the simulated wallet.

Day-by-Day Wallet Simulation



5. Day-by-Day Wallet Lifetime Simulation

```
function estimateDaysRemaining(state) {
  balance = state.walletBalance
  date = today
  cycleKWh = state.currentCycleKWh
  charged = state.chargedAmountForCurrentCycle
  days = 0

  while (balance > 0 && days < 365) {
    if (startsNewBillingCycle(date)) {
      cycleKWh = 0
      charged = 0
    }

    predictedKWh = usagePredictor.predict(date)
    projectedKWh = cycleKWh + predictedKWh

    newLiability = tariffEngine.calculateBill(projectedKWh, date)
    incrementalDebit = newLiability - charged

    balance -= incrementalDebit
    cycleKWh = projectedKWh
    charged = newLiability

    if (balance <= 0) return days + fractionOfDayIfNeeded()

    date = date + 1 day
    days++
  }

  return days
}
```

Billing-cycle reset

If the simulated date enters a new billing period, reset simulated cycle consumption and the simulated cumulative amount already charged for that cycle. Then continue with the tariff version applicable to the new date.

Tariff changes in the future

Tariffs should be versioned with effective dates. The simulation should request the tariff that is valid on each simulated date. This keeps long-running estimates correct even if a new tariff becomes active.

Fractional final day

For a more accurate display, when the wallet becomes insufficient during a simulated day, estimate the fraction of that day the remaining balance can cover. The UI may still round the result to a user-friendly value such as “~8 days”.

Conceptual example

```
Start: cycle usage = 82.00 kWh, wallet = LKR 3,970
Day 1: add predicted kWh -> calculate new cumulative tariff liability -> deduct only the increase
Day 2: repeat from the new simulated state
Continue across tariff thresholds and billing-cycle resets
Stop when the simulated wallet can no longer cover the next incremental liability
```

6. Backend Services and API Design

Keep prediction, tariff calculation, and presentation independent. This makes the system easier to test and prevents the frontend from becoming a second billing engine.

ConsumptionHistoryService	Build daily usage aggregates from smart-meter readings.
UsagePredictionService	Predict future kWh for a specific date.
TariffCalculationService	Return cumulative tariff liability for a cycle usage and date.
WalletLifetimeEstimator	Simulate future days until the wallet cannot cover the liability.
DashboardService	Return the estimate, range, confidence, and supporting values to the UI.

Suggested API response

```
{
  "walletBalance": 3970.00,
  "estimatedDaysRemaining": 8,
  "estimatedMinDays": 7,
  "estimatedMaxDays": 10,
  "averageDailyUsageKwh": 4.43,
  "predictedTomorrowKwh": 4.58,
  "billingCycleConsumptionKwh": 82.0,
  "predictionConfidence": "MEDIUM",
  "predictionMethod": "WEIGHTED_HISTORICAL",
  "calculatedAt": "2026-09-30T19:30:00+05:30"
}
```

When to recalculate

- A new valid smart-meter reading changes consumption.
- The user tops up the wallet.
- A new billing cycle starts.
- A new tariff version becomes effective.
- The stored usage pattern changes enough to materially alter the prediction.

7. Consumer Dashboard Presentation

The dashboard should communicate that the result is an estimate, not a guaranteed duration. Keep the main number simple and place uncertainty and supporting data underneath.

ESTIMATED DAYS REMAINING

~8 Days

Estimated range: 7-10 days

Based on your recent electricity usage.

Avg Daily Usage: 4.43 kWh/day

Prediction Confidence: Medium

Recommended tooltip

What does this mean?

This estimate uses your recent electricity usage, current wallet balance, billing-cycle consumption and the applicable electricity tariff. Actual duration can change if your future electricity usage changes.

Confidence

Start with a simple confidence label based on history length, then improve it later using consumption variability.

History	Confidence
< 7 reliable days	Low
7-29 reliable days	Medium
30+ reliable days	High

8. Validation and Edge Cases

The estimator is only trustworthy if it is tested at tariff boundaries, billing resets, and poor-data conditions.

Test area	Expected behavior
Tariff boundaries	Test exactly below, at, and above every tariff threshold, especially 60->61 and 180->181 kWh.
Billing reset	Verify a wallet estimate that crosses into the next billing period resets cycle kWh and prior cycle liability.
Fixed charge	Verify the same fixed charge is never deducted repeatedly for the same cumulative state.
Wallet top-up	After top-up, estimate should refresh using the new balance without changing historical usage.
No history	Return estimate unavailable rather than inventing a daily consumption value.
Missing/invalid meter data	Ignore duplicates, negative intervals, impossible decreases, and invalid timestamps according to meter rules.
Zero usage	Do not divide by zero. Return a safe state such as "Unable to estimate while no usage is detected".
Scenario ordering	Ensure minDays <= normalDays <= maxDays.
Money precision	Use decimal-safe monetary arithmetic and consistent rounding rules.

9. Future Improvement - Machine Learning

Only after the platform has enough household history should the prediction component be upgraded. Keep the tariff engine unchanged.

- Candidate features: hour/day, weekday/weekend, previous 24h usage, 7-day trend, previous same-day usage, month/season, and optional weather.
- Start with gradient-boosted tree models such as XGBoost or LightGBM before considering more complex sequence models.
- Evaluate prediction accuracy against the historical weighted-average baseline. Do not deploy a complex model unless it provides measurable improvement.
- Always predict kWh first; never train the model to reproduce tariff cost.

Final design rule

One production tariff engine should be the source of truth for real billing and future simulation. The predictor estimates kWh; the simulator advances through future dates; the tariff engine prices each cumulative state; the wallet determines when supply is no longer affordable.